An Explicit Defining Polynomial in Twenty-Seven Variables

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Abstract

Under the specified normalized elliptic, modular, pole-realization and prime-pattern inputs, we construct integer polynomials in twenty-seven total variables whose rational input projection is exactly the integers. An explicit sufficient coefficient parameter is $L = 125001$. Three comparison scales control the two rational components of a sum of four elliptic abscissas, so the height argument uses only effective constants. Scalar and comparison divisibility conditions share three pole predicates. A clustered choice of the comparison primes permits one additional congruence on the common elliptic index and makes linear scalar normalization sufficient. Nine quartic roots and three quadratic roots are eliminated with one separating coordinate. An exact integer certificate verifies the finite exceptional set and every numerical inequality in the sufficient threshold.

1 Statement and arithmetic inputs

A polynomial $F \in \mathbb{Z}[X, Y_1, \dots, Y_m]$ defines $\mathbb{Z}$ over $\mathbb{Q}$ when

$$x \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists \mathbf{y} \in \mathbb{Q}^m : \ F(x, \mathbf{y}) = 0 \qquad (x \in \mathbb{Q}).$$

All variable counts include the input X.

Theorem 1.1. *Under the arithmetic inputs below there is a computable family*

$$F_L \in \mathbb{Z}[X, Y_1, \dots, Y_{26}], \qquad L \geq 3 \text{ odd},$$

given by a finite integer coefficient specification. Every odd $L \geq 125001$ defines $\mathbb{Z}$ over $\mathbb{Q}$. In particular, F_{125001} is a fully specified defining polynomial. The integer L is a fixed coefficient parameter and is not a quantified variable. The total degree is at most $858293628694086519829299 20 < 2^{87}$, uniformly in L. For $L = 125001$, the maximum absolute integer coefficient is less than $2^{2^{102}}$. No additional abc or Szpiro hypothesis is used.

Write h for the absolute logarithmic Weil height, and put

$$K = \mathbb{Q}(\sqrt{2}), \quad d = 3 - \sqrt{2}, \quad D = d^2 = 11 - 6\sqrt{2}, \quad E : y^2 = x^3 - Dx,$$

$$P = \left(\frac{57 - 28\sqrt{2}}{8}, \frac{-660 + 417\sqrt{2}}{32} \right).$$

The nonidentity automorphism of K is denoted by σ. The following four arithmetic inputs have the scope established in [2, 8].

1. $2E(K) = \mathbb{Z}P$, the point P is non-torsion, and $h(x([n]P)) \geq n^2 - 6$ for every nonzero integer n.

2. The full normalized compatible modular realization for the five contacts is the one in [3]. Its proved consequence is

$$h(q) \le HM_5(q)^{7/3}, \qquad H = 2^{221234}, \tag{1}$$

for every rational q outside $0, 1, 2401, -343/32$. If $q = u/v$ is reduced with $v > 0$, $M_5(q)$ is the product of primes $p > 76176$ having positive odd valuation in at least one of

$$v, \quad u, \quad u - v, \quad u - 2401v, \quad 32u + 343v.$$

The modular realization, including the normalized compatible representation used in its multiplicity theorem, is retained as an explicit arithmetic input.

3. For $b \neq 0$ and $B_0 = 17b$, the pole predicate is

$$\begin{aligned} p_0^2 &= B_0(1-e)(1+3e), \\ B_0 q_0^2 &= (s^2 - B_0 e(B_0 - b^2))(s^2 + 3B_0 e(B_0 - b^2)). \end{aligned} \tag{2}$$

A rational solution excludes negative odd valuations of b outside $\{2, 3, 17\}$. A solution with $p_0 q_0 s \neq 0$ exists whenever

$$\begin{gathered} 0 < b < 17, \qquad b \text{ has square reduced denominator}, \\ b \in \mathbb{Q}_2^{\times 2} \cap \mathbb{Q}_3^{\times 2} \cap \mathbb{Q}_{17}^{\times 2}, \qquad v_2(b) \ge 3, \quad v_3(b) \ge 1, \quad v_{17}(b) \ge 2. \end{gathered} \tag{3}$$

The restricted-converse and descent inputs supporting this assertion are those identified in [2, 5].

4. The simultaneous prime-pattern input applies to any fixed finite list of rational contact coefficients. It supplies positive prime numerators, the prescribed Gaussian generators and conjugate reduction orders, arbitrary fixed denominator divisibility, arbitrary fixed avoidance, and arbitrarily large real slope. Its precise consequence of Sun's theorem is stated in [2].

The effective abscissa-sum estimate in [11] is proved from the first input. We will use it explicitly in Section 4. No ineffective aggregate trace-height bound enters the argument.

2 The rational source

Every displayed rational expression is restricted to its domain. All denominator guards are retained when clearing fractions. We also require the comparison slope λ and every pole input to be nonzero.

2.1 Nine elliptic points

Retain the nine points

$$A, S, T_1, Z, H_0, U_A, U_S, U_T, U_Z \in 2E(K) \setminus \{O\}.$$

For each point use two rational coordinates h_0, h_1 and the finite chart

$$h = h_0 + h_1\sqrt{2}, \qquad x = h^2, \qquad j^2 = h^4 - D, \qquad y = hj.$$

Under the group input these are precisely the nonzero multiples of P. Indeed the two-isogeny descent map shows that the square-abscissa subgroup is $2E(K)$; the squareclass and sign checks for the displayed curve are proved in [9].

For clarity, the one-root quartic chart for j is as follows. Write

$$A_0 = h_0^4 + 12h_0^2h_1^2 + 4h_1^4 - 11, \qquad B_0 = 4h_0h_1(h_0^2 + 2h_1^2) + 6,$$

$$c_0 = 2A_0 + B_0, \quad d_0 = -A_0 - B_0,$$

$$A' = 6A_0^3 - 9A_0B_0^2 - 4B_0^3, \qquad B' = -4A_0^3 + 6A_0B_0^2 + 3B_0^3.$$

Adjoin one root Y of

$$Y^4 - 4A'Y^2 + 8(B')^2 = 0, \qquad j = \frac{Y/2 + (B'/Y)\sqrt{2}}{c_0 + d_0\sqrt{2}}. \tag{4}$$

This chart is complete and separable at every rational retained-base specialization [6, 10]. Its discriminant is

$$2^{15}B'^2(2A_0^2 - B_0^2)^4(A_0^2 - 2B_0^2)^2 \neq 0.$$

The nonvanishing follows from the norm obstruction to $h^4 = D$, the identity $A'^2 - 2B'^2 = (2A_0^2 - B_0^2)^2(A_0^2 - 2B_0^2)$, and the Eisenstein cubic $4T^3 - 6T - 3$ excluding $B' = 0$.

Write the integer labels of (A, S, T_1, Z) as (n, a, t, b), and set

$$c = |a - 1|, \qquad B = \max(|n|, |a|, |t|, |b|).$$

The affine formula for $S - P$ includes its equal-abscissa guard, so $a \neq \pm 1$ and $c > 0$. Put

$$Q_0 = P, \quad (Q_1, Q_2, Q_3, Q_4) = (A, S, T_1, Z), \quad U_0 = H_0,$$

$$(U_1, U_2, U_3, U_4) = (U_A, U_S, U_T, U_Z), \qquad V_i = U_i - Q_i.$$

All differences use the usual guarded affine addition formulas.

2.2 Comparison and scalar functions

Write $f(Q) = 1/x(Q)$ and $z(Q) = -x(Q)/y(Q)$. The order-four formal-logarithm truncation is $z(Q)$ itself. Set

$$L_i = \frac{z(U_i)}{z(H_0)}, \qquad L_i' = \sigma\left(\frac{z(V_i)}{z(V_0)}\right) \quad (1 \le i \le 4),$$

and

$$\mathcal{F}(u, v, w) = v^2 - u(u-2)vw + (u-2)^2w^2 - 1, \qquad \beta = 216X^4 + 450459.$$

The sixteen comparison functions, in order, are

$$\begin{gathered} (f(U_i))_{i=0}^4, \qquad (\sigma f(V_i))_{i=0}^4, \\ (L_i - L_i')_{i=1}^4, \qquad L_1 - \beta, \qquad \mathcal{F}(L_2, L_3, L_4). \end{gathered} \tag{5}$$

Denote them by $C_1, \ldots, C_{16}$, and put

$$m_i = 1 \quad (i \in \{1, \ldots, 11, 15\}), \qquad m_i = 3 \quad (i \in \{12, 13, 14, 16\}).$$

Let $\eta_1, \ldots, \eta_6 \in \mathbb{Q}$ be the two rational coefficients in the basis $1, \sqrt{2}$ of each of

$$f(H_0), \qquad f(S - P), \qquad z(S - P)/z(H_0), \tag{6}$$

in that order. Assign the scalar orders

$$b_k = 1 \quad (1 \le k \le 4), \qquad b_k = 3 \quad (k = 5, 6).$$

For fixed odd $L \geq 3$, define

$$\ell = 76249, \qquad r = \beta(\beta-1)(\beta-2401)(32\beta+343)(2\beta-1),$$
$$C_L = 17(\ell^{(L+1)/2}+1), \qquad \rho = \frac{102^2 \ell^L r}{(1+C_L r)^2}. \tag{7}$$

The scalar normalization theorem [4] gives

$$v_\ell(\rho) = \begin{cases} L, & v_\ell(X) \geq 0, \\ L - 20v_\ell(X), & v_\ell(X) < 0. \end{cases} \tag{8}$$

This valuation is positive and odd. The same theorem gives, for real X,

$$0 < \rho < \frac{36}{\ell r} < 1. \tag{9}$$

For integral X, the reduced denominator of ρ is a square; its squareclasses and depths at $2, 3, 17$ satisfy (3), and $v_2(\rho) = 4$.

2.3 Three scales and three pole inputs

Put

$$\Theta = \{0, -1/2, -1, 1/2399, -32/407\},$$
$$\Theta_j = (4j+2+\Theta) \cup (4j+4+\Theta), \qquad \gamma_j = -j-1 \quad (j = 0, 1, 2).$$

All thirty contacts are positive. The three contact-and-pole sets are pairwise disjoint. Write

$$\mathcal{S} = \sum_{i=1}^{4} x(Q_i) = S_0 + S_1\sqrt{2}, \qquad U = S_0, \qquad V = S_0 + S_1,$$

and define

$$A_0 = 1, \qquad A_1 = (1+\rho U^2)^2, \qquad A_2 = (1+\rho V^2)^2, \qquad t_j = A_j \lambda. \tag{10}$$

These scales are positive rational squares. For $j = 0, 1, 2$, set

$$\delta_j = t_j - \gamma_j, \qquad R_j = \frac{\prod_{\theta \in \Theta_j}(t_j - \theta)}{\delta_j^{10}}, \qquad w_j = \rho R_j. \tag{11}$$

Require every contact and δ_j to be nonzero.

For nonzero rational q put $\Phi_q(v) = qv/(1+qv)$ on its nonzero-denominator domain. Define

$$G_{j,k} = R_j^{b_k} \Phi_{R_j}(\eta_k), \tag{12}$$

$$W_j = \prod_{i=1}^{16} \left(1 + w_j \left(\frac{C_i}{\delta_j w_j^{m_i}}\right)^2\right) \prod_{k=1}^{6} \left(1 + w_j \left(\frac{G_{j,k}}{w_j^{b_k}}\right)^2\right), \tag{13}$$

$$b_j = \frac{4\rho R_j}{\mathrm{Tr}_{K/\mathbb{Q}}(W_j^{-1})^2}. \tag{14}$$

Each b_j receives the pole predicate (2). Its first conic uses two retained rational parameters τ_j, s_j and

$$e_j = \frac{\tau_j^2 - B_j}{\tau_j^2 + 3B_j}, \qquad p_j = \frac{-4B_j\tau_j}{\tau_j^2 + 3B_j}, \qquad B_j = 17b_j.$$

The second equation is a single nonzero quadratic root after rational square clearing. The nonzero-witness form of the inherited converse makes this conic chart complete [5].

Let Σ be the primes at most 76176, together with every prime dividing a numerator or denominator of a contact, a pole, or a nonzero difference between two such values. Exact factorization gives

$$\Sigma \setminus \{p : p \le 76176\} = \{149027, 272011, 1875611, 3828397, 3982747\}. \tag{15}$$

Their product, denoted by E_*, satisfies $E_* < 2^{100}$, and $\ell \notin \Sigma$. The checker in Section 8 verifies these assertions independently by integer trial division.

3 Local divisibility

At primes outside Σ all valuations on completions of K are integer valued and normalized by $v(p) = 1$. A zero test function has valuation $+\infty$.

Lemma 3.1. *Suppose $\alpha = v_p(w_j)$ is odd and b_j has a pole certificate. Then $\alpha > 0$. At some place of K above p, every factor of W_j has valuation greater than $-\alpha/2$.*

Proof. The nonzero second summand of a factor $1 + w_j q^2$ has odd valuation, so that factor has valuation at most zero. A zero second summand gives the unit factor 1. Hence W_j^{-1} is integral at every place above p, and its trace is integral. The pole obstruction and (14) give

$$0 \le \alpha - 2v_p(\mathrm{Tr}(W_j^{-1})), \qquad v_p(\mathrm{Tr}(W_j^{-1})) \le (\alpha - 1)/2.$$

Thus $\alpha > 0$. At least one local value of W_j^{-1} has valuation at most $(\alpha - 1)/2$; otherwise so would its trace. At that place the sum of the nonpositive valuations of the factors of W_j is greater than $-\alpha/2$. Each summand is therefore greater than $-\alpha/2$. □

Lemma 3.2. *If $v_p(\rho)$ is odd, then at least one of R_0, R_1 is a unit.*

Proof. A balanced function R_j is a unit whenever $v_p(t_j) \neq 0$, because all contacts and poles are units. If $U = 0$, then $A_1 = 1$, and disjointness of the two reduced contact-and-pole sets proves the assertion directly. Otherwise write $\kappa = v_p(\rho U^2)$, which is odd. If $\kappa > 0$, then $A_1 \equiv 1 \pmod p$. For nonunit λ, both arguments are nonunits and both contact functions are units. For unit λ, the arguments have the same residue; the disjointness of the two contact-and-pole sets modulo p prevents both functions from being nonunits. If $\kappa < 0$, then $v_p(A_1) = 2\kappa \neq 0$. The arguments t_0 and t_1 cannot both be units, so at least one of their balanced functions is a unit. □

Lemma 3.3. *If $\mu = v_p(\rho)$ is odd, then $\mu > 0$ and*

$$v_p(c) \ge 2\mu + 1 \ge 3.$$

In particular

$$c \ge \ell^{2L+1}, \qquad 4L \log \ell \le 2 \log c. \tag{16}$$

Proof. Choose a unit R_j by Lemma 3.2. Lemma 3.1 forces $\mu > 0$. Applied to each scalar factor, it gives

$$v(G_{j,k}) > (b_k - 3/4)\mu.$$

For a unit R_j, positive valuation of $\Phi_{R_j}(\eta_k)$ implies positive valuation of η_k, and the two valuations are equal. Thus the first four rational scalar components have valuation $> \mu/4$, while the last two have valuation $> 9\mu/4$. The first four put both H_0 and $S - P$ in the formal group at every place above p. If $H_0 = [h]P$, the full formal-logarithm ratio is $(a - 1)/h$. The

quotient of a formal parameter and its logarithm is a unit. The third pair of scalar components consequently gives

$$v_p(a-1) - v_p(h) = v(z(S-P)/z(H_0)) \geq 2\mu + 1.$$

Since h is an integer, this proves the asserted bound for c. Apply it at ℓ, using (8), to obtain (16). □

Lemma 3.4. *Let $v_p(R_j) = \nu > 0$ be odd and suppose $v_p(\rho)$ is even. At some place above p, every function in (5) has valuation at least its assigned order m_i.*

Proof. At such a contact prime δ_j is a unit. The integer $\alpha = v_p(\rho) + \nu$ is odd. By Lemma 3.1, $\alpha > 0$ and the comparison factors at one place satisfy

$$\alpha + 2\big(v(C_i) - m_i\alpha\big) > -\alpha/2$$

whenever the second summand is not integral. The same resulting lower bound holds if it is integral. Thus

$$v(C_i) > (m_i - 3/4)\alpha \geq m_i - 3/4,$$

which gives $v(C_i) \geq m_i$ by integer rounding. □

For a point in the first formal neighborhood at $p > 76176$, one has

$$v(\log_E(Q) - z(Q)) \geq 5v(z(Q)), \qquad z(Q)/\log_E(Q) \in 1 + p^4\mathcal{O}. \tag{17}$$

This follows from the formal logarithm of $y^2 = x^3 - Dx$, whose first omitted term has parameter order five; the integral-denominator estimate at these primes is proved in [5, 7]. Positive valuation of $1/x(Q)$ is equivalent to membership in the first formal neighborhood.

Suppose Lemma 3.4 applies, and write $U_i = [u_i]P$, $H_0 = [h]P$, and $(n_1, n_2, n_3, n_4) = (n, a, t, b)$. The exact logarithm ratios are

$$r_i = u_i/h, \qquad r_i' = (u_i - n_i)/(h-1).$$

At least one of $h, h-1$ is a p-unit, so its ratio is integral. The cross comparison and (17) make the other truncated ratio integral as well. Both relative errors are then absolute errors. One has $L_1 \equiv r_1 \equiv r_1' \pmod{p}$, while $L_i \equiv r_i \equiv r_i' \pmod{p^3}$ for $i = 2, 3, 4$. Multiplying by $h, h-1$ and subtracting gives

$$L_1 \equiv n \pmod{p}, \qquad L_i \equiv n_i \pmod{p^3} \quad (i = 2, 3, 4).$$

Define the integer residual $\Delta = \mathcal{F}(a, t, b)$. If $\beta = u/v$ is reduced with $v > 0$, the direct target also makes β integral at p. Consequently

$$p^3 \mid \Delta, \qquad p \mid u - nv.$$

Let M_j be the product of primes outside Σ at which R_j has positive odd valuation. At a prime in M_j, an odd scalar valuation invokes Lemma 3.3; an even scalar valuation invokes Lemma 3.4. Consequently

$$M_j^3 \mid \Delta c \quad (\Delta \neq 0), \qquad M_j \mid (u - nv)c \quad (u - nv \neq 0), \qquad j = 0, 1, 2. \tag{18}$$

No prime shared by the scalar and comparison coefficients is omitted.

4 Effective height bounds

Set

$$\overline{H} = 2^{221470}. \tag{19}$$

All constants in this section are independent of X and L.

4.1 Scalar contacts and a coprime divisibility

Let M_β° be the product of primes in $M_5(\beta)$ outside Σ. Such a prime has positive odd valuation in ρ. Indeed, a negative valuation of β is a multiple of four, so its denominator cannot contribute an odd contact. At a finite contact exactly one of the five factors of r has positive odd valuation, and $1 + C_L r$ is a unit. The additional factor $2\beta - 1$ is a unit at these four finite contacts outside Σ. Lemma 3.3 gives

$$(M_\beta^\circ)^3 \mid c.$$

The prime ℓ does not contribute to $M_5(\beta)$. For ℓ-integral X, enumeration of fourth-power residues excludes $\beta \equiv 0, 1, 2401, -343/32 \pmod{\ell}$; for nonintegral X, $v_\ell(\beta)$ is a multiple of four. The independent certificate in Section 8 checks the finite enumeration. Combining this coprimality with (16) yields the stronger product divisibility

$$\ell^{2L+1}(M_\beta^\circ)^3 \mid c. \tag{20}$$

Removing the five additional exceptional primes from a contact radical changes it by a factor at most E_*. Therefore (1) and $E_* < 2^{100}$ give

$$h(\beta) \le \overline{H}(M_\beta^\circ)^{7/3} \le \overline{H}\left(\frac{c}{\ell^{2L+1}}\right)^{7/9} \le \frac{\overline{H}}{\ell^{2L+1}}c. \tag{21}$$

The last inequality uses $c/\ell^{2L+1} \ge 1$. The factor ℓ^{2L+1} is retained before applying the height bound; it is not absorbed into an unspecified constant.

The scalar coefficient also has the elementary estimate

$$h(\rho) \le 15h(\beta) + 2L\log\ell + 128. \tag{22}$$

To prove it, put $K_0 = 2 \cdot 2402 \cdot 375 \cdot 3 = 5404500$. The elementary height inequalities give

$$h(r) \le 5h(\beta) + \log K_0,$$

$$h(\rho) \le \log 102^2 + L\log\ell + 3h(r) + 2\log C_L + 2\log 2.$$

Since $C_L \le 34\ell^{(L+1)/2}$, the remaining constant is

$$3\log K_0 + 2\log 34 + \log\ell + \log 102^2 + 2\log 2 < 114 < 128.$$

The displayed bound 114 follows already from the binary lengths of the five positive integers and $\log 2 < 1$.

4.2 Ten-contact heights

For $s \in \{2, 4, 6, 8, 10, 12\}$ the rational map

$$q_s = 2 + \frac{1}{t - s}$$

sends $s+\Theta$ to the five contacts $\infty, 0, 1, 2401, -343/32$. Its inverse has the integral matrix

$$\begin{pmatrix} s & 1-2s \\ 1 & -2 \end{pmatrix},$$

whose determinant is -1 and whose largest absolute row sum is at most 35. Thus $h(t) \le h(q_s) + \log 35$. Outside Σ, this map preserves contact multiplicities, and the odd radical of q_s is exactly the odd radical of the five translated contacts. Denote that radical by M_s.

For each j, the two five-contact sets are disjoint modulo every prime outside Σ. The denominator δ_j^{10} contributes only even valuations. Hence

$$M_{4j+2}M_{4j+4} = M_j.$$

If $h(t_j) > \log 35$, applying (1) to both five-contact sets and multiplying gives

$$\bigl(h(t_j) - \log 35\bigr)^2 \le H^2 E_*^{14/3} M_j^{7/3}.$$

The same final bound is immediate if $h(t_j) \le \log 35$. Since

$$\log 35 + H E_*^{7/3} < 2^{221470} = \overline{H},$$

we conclude that

$$h(t_j) \le \overline{H} M_j^{7/6} \qquad (j = 0, 1, 2). \tag{23}$$

All excluded primes and the height cost of translation have now been included explicitly.

4.3 Two rational components of the abscissa sum

The effective sum-height theorem [11] gives

$$B^2 \le 2h(\mathcal{S}) + 80.$$

The variables $U = S_0$ and $V = S_0 + S_1$ satisfy

$$\mathcal{S} = (1-\sqrt{2})U + \sqrt{2}V.$$

At finite places both coefficients are integral. The product of the two real coefficient sums is $(2\sqrt{2}-1)(2\sqrt{2}+1) = 7$. It follows that

$$h(\mathcal{S}) \le h(U) + h(V) + \tfrac{1}{2}\log 7, \qquad B^2 \le 2h(U) + 2h(V) + 82.$$

This uses $\log 7 < 2$.

For $W = U, V$, respectively, the two nontrivial scales satisfy

$$h((1+\rho W^2)^2) \ge 4h(W) - 2h(\rho) - 2\log 2.$$

The nonzero slope gives $A_j = t_j/t_0$. Combining these inequalities with $2\log 2 < 2$ yields

$$B^2 \le h(t_0) + \tfrac{1}{2}h(t_1) + \tfrac{1}{2}h(t_2) + 2h(\rho) + 84. \tag{24}$$

Consequently, (18), (22), and (23) give the effective common inequality

$$B^2 \le 2\overline{H}|Nc|^{7/(6k)} + 30h(\beta) + 4L\log\ell + 340, \tag{25}$$

where $(N,k) = (\Delta, 3)$ if $\Delta \neq 0$, and $(N,k) = (u-nv, 1)$ if $u - nv \neq 0$.

5 An explicit sufficient coefficient parameter

Assume throughout this section that $L \geq 125001$ is odd. As $\ell > 2^{16}$, equation (16) gives

$$c \geq \ell^{2L+1} > 2^{4000048}. \tag{26}$$

In particular $\ell^{2L+1} > 120\overline{H}$, so (21) implies

$$h(\beta) \leq c/120. \tag{27}$$

Using $4L \log \ell \leq 2 \log c$, the last three terms of (25) are bounded above by

$$c/4 + 2 \log c + 340.$$

For $c \geq 100$, this is less than $(c-1)^2/2 \leq B^2/2$. For example, use $\log c \leq c$: the polynomial $(c-1)^2/2 - 9c/4 - 340$ is positive at 100 and increasing thereafter. Thus these three terms can be absorbed into $B^2/2$.

5.1 Vanishing of the Lucas residual

If $\Delta \neq 0$, the elementary bounds $B > 4$, $c \leq 2B$ and $|\Delta| \leq 5B^4$ give $|\Delta c| \leq 10B^5$. Equation (25) then implies

$$B^{1/18} \leq 4\overline{H}10^{7/18}, \qquad B \leq 4^{18}10^{7}\overline{H}^{18} < 2^{3986520}.$$

Here $10^7 < 2^{24}$ gives the last strict inequality. This contradicts $B \geq c - 1$ and (26). Therefore

$$\mathcal{F}(a, t, b) = 0. \tag{28}$$

We recall the elementary growth consequence of this equation, including its proof. If $|a| > 7$ and $b \neq 0$, then

$$\log B > c - 4. \tag{29}$$

Indeed, put $A = |a|$ and $C = |(a-2)b|$. Equation (28) implies that $T = |t|$ and C are positive integer solutions of $T^2 - ATC + C^2 = 1$. Their signs before taking absolute values must agree after multiplying $(a-2)b$ by $\operatorname{sgn}(a)$. For the recurrence $U_0 = 0$, $U_1 = 1$, $U_{j+1} = AU_j - U_{j-1}$, every positive solution is an unordered pair of adjacent terms. To see this, order its coordinates $M \geq m > 0$ and replace them by

$$(m, Am - M), \qquad Am - M = (m^2 - 1)/M \in [0, m).$$

This preserves the equation and descends to $(1, 0)$. Hence $C = U_j$ for some $j \geq 1$ and $T \geq U_{j-1}$. Writing $d = |a - 2|$, the recurrence modulo d gives $U_j \equiv j$ if $a > 0$ and $U_j \equiv (-1)^{j-1}j$ if $a < 0$. Since $d \mid C$, we have $j \geq d \geq A - 2$. Also $U_r > (A-1)^{r-1}$ for $r \geq 2$, by induction. It follows that

$$\log T > (j-2)\log(A-1) \geq (A-4)\log 7 > A - 3 \geq c - 4,$$

proving (29). Its hypotheses hold here because c satisfies (26) and all root labels are nonzero.

5.2 Integrality of the input

Suppose $\beta \neq n$. For $\beta = u/v$ reduced with $v > 0$, put $N = u - nv \neq 0$. Then

$$|N| \leq 2e^{h(\beta)}B.$$

Retain c separately in the product $|Nc|$. After the same absorption, equation (25), now with $k = 1$, gives

$$B^{5/6} \leq 4\overline{H}2^{7/6}e^{(7/6)h(\beta)}c^{7/6}.$$

Taking logarithms and using $\log 2 < 1$ yields

$$\log B < 265768 + \frac{7}{5}\big(h(\beta) + \log c\big) \le 265768 + \frac{217}{600}c < c - 4. \tag{30}$$

Here $h(\beta) \le c/120$ and $\log c \le c/4$ for $c \ge 16$. The last inequality holds for $c \ge 10^6$: it is strict at 10^6 and the coefficient $217/600$ is less than one. This contradicts (29), so $\beta = n \in \mathbb{Z}$. If $X = u_0/v_0$ is reduced with $v_0 > 0$, integrality of $216X^4 + 450459$ implies $v_0^4 \mid 216 = 2^3 3^3$. Thus $v_0 = 1$ and $X \in \mathbb{Z}$. This proves soundness for every fixed odd $L \ge 125001$.

6 Completeness

Fix an integer input X and any fixed odd $L \ge 3$. The construction below gives a rational source witness. Its order of choices is relevant: the prime-pattern input is applied only after the root labels, scales, neighborhoods and finite avoidance conditions have been fixed.

6.1 Scalar preparation and fixed neighborhoods

Let $\mathcal{P}$ consist of the finite prime support of ρ, Σ, and $2, 3, 17$. Choose $H_0 = [h_0]P$, with positive $h_0 > 2$ divisible by 64, in sufficiently deep formal subgroups at the finite preparation places and sufficiently close to the identity at both real embeddings. Such real returns exist in the compact closure of the cyclic subgroup after any finite index restriction. Now choose a positive c divisible by 6 and any additional required finite modulus, with $[c]P$ still closer to the identity at those places. Put $a = c + 1$ and $S = [a]P$. The six scalar components in (6) can thereby be made arbitrarily small simultaneously. For the ratio, H_0 has already been fixed with nonzero local parameter, and $[c]P$ can approach the identity as closely as needed. Deep formal subgroups and continuity give this assertion also at ramified and bad primes, as in the scalar realization of [8].

With ρ fixed, make the scalar factors in (13) arbitrarily close to 1 whenever R_j is close to 1 at a finite preparation prime, and whenever $1/2 < R_j < 2$ over the reals. In the latter case also require $|\eta_k| < 1/4$, so $1 + R_j\eta_k > 1/2$ for all six denominators.

Set $n = \beta$. Define

$$U_0 = 0, \quad U_1 = 1, \quad U_{m+1} = aU_m - U_{m-1}, \qquad k = 2(a-2), \quad b = U_k/(a-2), \quad t = U_{k+1}.$$

The recurrence modulo $a - 2$ gives $U_k \equiv k \equiv 0$, so b is integral. Its usual determinant identity gives $\mathcal{F}(a, t, b) = 0$. Since a is odd and $a \equiv 1 \pmod 3$, the sequence modulo two has period $0, 1, 1$ and $k \equiv 1 \pmod 3$. Thus n, a, t, b are positive and odd. The root points and all three positive square scales are now fixed.

Temporarily use $H = [h]P$, $U_i = [hn_i]P$. Restrict h to a fixed congruence class at the finite preparation places and to a compact real neighborhood preserving all scalar bounds. Shrink these neighborhoods so that all raw comparison functions remain bounded and all their original charts are defined. The relevant multiplication and translation maps have only finitely many poles; at an integer $h > 2$ their equal-abscissa failures are absent because $n_i \ne 0$. There are no comparison normalization denominators. Take the initial modulus to be divisible by 64 and by the order of the component group of the compact real closure of $\mathbb{Z}P$. That component group is finite, since the closure is a compact real Lie subgroup.

6.2 A clustered prime choice and the common index

Apply the rational-contact version of the prime-pattern input to $-\theta/A_j$, for $j = 0, 1, 2$ and $\theta \in \Theta_j$. Repeated coefficients are listed once, with all incidences retained. It provides

$$\lambda = u/v, \qquad v = D_0^2 w^2,$$

with every $p_i = u - (\theta/A_j)v$ a prescribed positive prime. Require these primes to avoid every fixed preparation prime, every numerator or denominator of a scale, every relevant contact difference, and all primes where a coordinate or the nonzero local parameter of P or $S - P$ is not a unit. These are finite conditions, fixed before prime selection. Prescribe enough divisibility of w to make λ arbitrarily large in absolute value at all finite preparation places, and choose it arbitrarily large and positive at the real place. The rational-contact and avoidance consequences are those established in [2].

As λ tends to infinity at any preparation place, R_j tends to 1 and δ_j^{-1} to zero. The scalar factors are close to 1 by preparation. The comparison factors tend uniformly to 1 on the bounded comparison neighborhoods. Thus all required finite and real closeness conditions can be imposed by fixed requirements before invoking the prime-pattern input.

Impose additionally

$$\lambda > 4 \max_{j,\theta\in\Theta_j} (\theta/A_j), \qquad u \geq 25. \tag{31}$$

All contacts are positive, so every selected prime lies in $(3u/4, u)$. The full order of either conjugate reduction of E at any selected prime p_k is even and bounded by

$$p_k + 1 + 2\sqrt{p_k} < u + 1 + 2\sqrt{u} < 3u/2.$$

The evenness follows from the rational two-torsion. For $u \geq 25$, $\sqrt{u} \leq u/5$ and $1 < u/10$, proving the last inequality. If a selected odd prime p_i divided one of these even group orders, that order would be at least $2p_i > 3u/2$, a contradiction. Consequently every selected prime is coprime to every prescribed annihilating reduction modulus.

The inherited conjugate-order construction imposes the primary annihilation on h, the opposite annihilation on $h - 1$, and the initial fixed congruence on h. Its odd coprimality and two-adic compatibility are proved in [2]. The preceding paragraph permits the additional simultaneous congruences

$$h \equiv 0 \pmod{\prod_i p_i}. \tag{32}$$

All selected primes also avoid the initial fixed modulus.

This refined arithmetic progression still meets the prepared real neighborhood. Indeed, let G be the compact closure of $\mathbb{Z}P$ and G° its identity component. If the initial modulus m annihilates G/G°, then $mG = G^\circ$: containment is clear, and multiplication by m on the compact torus G° is surjective. For every positive integer r one likewise has $mrG = G^\circ$. Hence every compatible refinement of the index congruence has the same closure in the prepared component. Choose an index $h > 2$ in the refined progression and real neighborhood. Set $H_0 = [h]P$ and $U_i = [hn_i]P$.

6.3 The linear scalar tests at the selected primes

At a selected prime p for detector j, one has $v_p(R_j) = 1$, while ρ and δ_j are units. The reduction-order conditions put the required U_i and V_i in their first formal neighborhoods. The first ten comparison functions therefore have valuation at least two. Their exact logarithm ratios are the integers n_i; equation (17) gives valuation at least four for the last six comparisons. At the designated place every comparison second summand in (13) thus has valuation at least three.

We verify separately the six linear scalar denominators. At the primary place, $[h/p]P$ is in the formal group: division by p preserves reduction annihilation because the reduction modulus is coprime to p. Since $H_0 = [p]([h/p]P)$, its local parameter has valuation at least two. At the opposite place, H_0 reduces to P, whose coordinates and parameter are units by avoidance. The selected primes split in K, as provided by the prime pattern. The two rational components of $f(H_0)$ are units: its two local values have respectively positive and

zero valuation. The two components of $f(S-P)$ are integral. Finally, the local values of $z(S-P)/z(H_0)$ have valuations at most -2 and exactly zero; its two rational components consequently both have valuation at most -2. Here 2 and $\sqrt{2}$ are units, so the sum and scaled difference of two values with unequal valuations preserve the smaller valuation.

Every rational component η_k thus has either nonnegative valuation or valuation at most -2. Since $v_p(R_j) = 1$, $v_p(R_j\eta_k) \neq 0$. The identity

$$v\left(\frac{x}{1+x}\right) = \max(0, v(x)) \quad (v(x) \neq 0)$$

shows that every $\Phi_{R_j}(\eta_k)$ is integral. Consequently $v_p(G_{j,k}) \geq b_k$, and all scalar second summands in (13) have positive valuation at both places over p. No upper bound on the initial formal depth is used.

6.4 The pole certificates

The exact contact factorization is

$$R_j = \left(\frac{A_j}{v\delta_j}\right)^{10} \prod_{\theta\in\Theta_j} (u - (\theta/A_j)v)\,.$$

Thus R_j is a positive rational square times its ten selected primes, each of which has valuation exactly one and avoids the support of ρ. At the designated place over such a prime, $W_j^{-1} \equiv 1$. At the conjugate place every factor has nonpositive valuation, since a nonzero second summand has odd valuation. Its inverse is congruent to zero or one, and so $W_j^{-1} \equiv 0$ or 1 there. The trace is therefore congruent to 1 or 2 and is a unit. The odd valuation of b_j is positive. At primes in the support of ρ, both R_j and W_j are arbitrarily close to 1 by preparation, so the odd valuations remain nonnegative in b_j. At every other prime the valuation of ρR_j is even, as is its change under division by the square trace. Each b_j has square reduced denominator.

At $2, 3, 17$, choose the same neighborhoods so that R_j is a local square and $\mathrm{Tr}(W_j^{-1})$ is sufficiently close to 2. The squareclass of b_j is then that of ρ. At two, the factor 4 in (14) cancels the limiting trace square and gives $v_2(b_j) = 4$; the required depths at three and seventeen are unchanged. At the real embeddings arrange $1/2 < R_j < 2$ and $1 \leq W_j < 2$. Every second summand is positive or zero, and preparation makes their finite product less than two. The trace is at least one, giving

$$0 < b_j < 8\rho < 17.$$

All scalar denominators are nonzero by $|R_j\eta_k| < 1/2$; all products and traces are nonzero by positivity. The inherited converse now supplies the three nonzero pole witnesses in the specified conic charts. All nine point charts are available, since the root labels are nonzero and the guarded differences have $h > 2$, $a > 1$. This completes the source witness for every integer input and every fixed odd $L \geq 3$.

7 One polynomial and its numerical bounds

The retained coordinates consist of the input X, eighteen point coordinates, one nonzero slope, and six pole parameters. Their number is twenty-six. The finite algebra has nine quartic generators and three nonzero quadratic pole generators, hence rank

$$4^9 2^3 = 2^{21}.$$

The active finite-algebra construction [1] eliminates all twelve finite roots and retains all guards with one additional rational coordinate. This gives exactly twenty-seven total variables.

Here is its finite polynomial specification. Let B_0 be the algebra of the nine point roots and A_0 the algebra after adjoining the three pole roots q_1, q_2, q_3, with $q_j^2 = c_j \in B_0$ after rational square clearing. Put $b_0 = 2^{18}$, $n_0 = 2^{21}$. The complete guard has the form $G = gq_1q_2q_3$, with $g \in B_0$. There are no additional closed equations. Define

$$T_0 = g^2 c_1 c_2 c_3, \qquad \det_{B_0}(I - vM_{T_0}) = \sum_{j=0}^{b_0} a_j v^j, \qquad a_0 = 1.$$

Let $\theta_1, \ldots, \theta_m$, with $m = 12$, be the point and pole generators, and let δ_i be the discriminant of point quartic i. All nine discriminants are nonzero at every rational base specialization. Set

$$R = n_0 \left(1 + \sum_{i=1}^{m} \sum_{u,v=1}^{n_0} (M_{\theta_i}^{A_0})_{uv}^2 + \sum_{u,v=1}^{b_0} (M_g^{B_0})_{uv}^2 \right).$$

For $1 \le r \le b_0$, define

$$c_r = a_r, \qquad h_r(Z) = \sum_{j=0}^{r} a_j Z^{r-j}, \qquad E_r = c_r - h_r(T_0),$$

$$D_r = c_r^2 \prod_{i=1}^{9} \delta_i^2, \qquad s_r = 4r + 108,$$

$$K_r = (2R)^{s_r}(1 + c_r^2) \prod_{i=1}^{9} (1 + \delta_i^2), \qquad C_r = D_r + 4RK_r,$$

$$\beta_r = D_r E_r \left(C_r^m + \sum_{i=1}^{m} D_r^i C_r^{m-i} \theta_i \right), \qquad H_r(U) = \det_{A_0}(I - UM_{\beta_r}).$$

Then

$$F_L = H_{b_0}(U) \prod_{r=1}^{b_0 - 1} \left(H_r(U)^2 + \sum_{j=r+1}^{b_0} a_j^2 \right) \tag{33}$$

is the required polynomial after clearing the fixed integer coefficient denominators in the prescribed source construction. The active finite-algebra theorem proves that its rational projection is exactly the guarded source projection. The displayed product has 262144 factors. It is a finite specification by integer polynomial operations; no expanded coefficient table of this very large polynomial is needed.

The independent weighted-degree calculation gives, uniformly in L,

$$\deg F_L \le 858293628694086519829299 20 < 2^{87}. \tag{34}$$

At the explicit defining member $L = 125001$, the maximum absolute integer coefficient is less than

$$2^{2^{102}}. \tag{35}$$

The separating variable has degree at most $E = 1099509530624$. Writing $D = 858293628694086519829299 20$, the support is therefore bounded above by

$$\binom{D+27}{27} - \binom{D-E+26}{27} < 2^{2192}.$$

This is a support upper bound, not an expanded monomial count. These bounds, together with the effective soundness and completeness proofs, establish Theorem 1.1.

8 Exact certificates

All verification programs use the Python standard library. The independent constant checker is

`computation/few_variables/effective27_v19/check_effective27_constants.py`.

It factors the numerators and denominators of all thirty contacts, the three poles and all pairwise differences. It verifies the five extra exceptional primes, their product

$$E_* = 1159294788412841845761453254053 < 2^{100},$$

the primality and admissibility of ℓ, all 76249 input residues modulo ℓ, and the strict numerical inequalities leading to (26), (30) and the contradiction for $\Delta \neq 0$. It uses only integers and exact rational arithmetic, including rational lower bounds for the exponential where a logarithmic comparison is needed. Its output is `effective27-constants-certificate.json`.

The separate source and finite-algebra certificates in the same module reconstruct the thirty contacts, three scales, sixteen raw comparison tests, six scalar components, three pole charts and every guard. The source has 6898 rational arithmetic nodes, 2588 integer coefficient nodes and 164 atomic guard factors. The source-to-integer conversion, quartic relations, square clearings and guard coverage are checked by exact symbolic identities. Fifteen independent reconstructions over three finite fields provide additional finite evidence for the rational expression formulas; they are not a proof of universal polynomial identities. The reports `independent-source-verification.json` and `independent-bound-verification.json` verify the stated source census, all 262144 degree cases, the degree and support bounds, and the coefficient bound at $L = 125001$. These finite checks certify the specification and inequalities stated here; the arithmetic inputs remain precisely those in Section 1.

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